

VIKHYAT GUPTA

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EDUCATION

Amity University Punjab

Bachelor of Technology - Computer Science

Sep 2024 – Sep 2028

Zirakpur, Punjab

PROJECTS

Real-Time Server-Based Communication Platform | Next.js, React, Node.js, MongoDB, Socket.IO

- Engineered a **full-stack, real-time communication system** inspired by Discord, supporting **multi-server, multi-channel messaging** for concurrent users.
- Designed and implemented **role-based access control (RBAC)** to securely manage permissions for server owners, admins, and members.
- Built **protected routes and authentication workflows**, preventing unauthorized access to servers and channels.
- Integrated **WebSocket-based real-time messaging** using Socket.IO, ensuring low-latency message delivery and synchronized chat state.
- Optimized data flow and UI rendering to support **scalable server and channel creation** with persistent chat history.

TruthFlow – AI-Powered Legal Testimony Structuring Platform | Next.js, FastAPI, MongoDB

- Engineered a **full-stack AI-powered legal documentation platform** that converts fragmented survivor testimony into **structured, court-ready reports** using LLM-based processing pipelines.
- Integrated a **FastAPI backend** with Next.js frontend, handling API orchestration, report generation workflows, incremental refinement, and secure file uploads.
- Built a **multi-stage AI refinement system** that generates contextual follow-up questions, preserves session continuity, and incrementally improves legal report accuracy.
- Designed structured **JSON-based report schemas** for timelines, people, locations, and evidence linkage using validation models.
- Implemented **draft persistence, evidence management, and conversational session flows** to support trauma-aware, non-linear testimony collection.
- Integrated **LLM APIs, secure backend handling, PDF report generation, and cloud-based evidence storage** for scalable legal-tech workflows.

ReviewSync | React, Node.js, Express, MongoDB, Socket.IO

- Developed a **collaborative code review platform** enabling **real-time multi-user code editing** and discussion.
- Implemented **live synchronization** using WebSockets (Socket.IO), handling concurrent edits with minimal latency.
- Designed a **secure RESTful backend** using Node.js, Express, MongoDB, and Passport for authentication and authorization.
- Integrated **Monaco Editor** for an in-browser IDE-like experience with syntax validation and form handling.
- Deployed on Render; analyzed **WebSocket limitations in production environments**, gaining experience with real-world deployment constraints.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL

Web & Frameworks: Next.js, React.js, Node.js, Express, FastAPI, Flask, Tailwind CSS

Databases & Storage: MongoDB, SQL, AWS S3

AI & Backend Systems: REST APIs, LLM Integration, Prompt Engineering, Pydantic, JSON Schema Design

Systems & Concepts: OOP, Authentication, Authorization, RBAC, WebSockets, Secure File Handling

Tools & Platforms: Git, GitHub, Socket.IO, Postman

CERTIFICATIONS

- CS50x: Introduction to Computer Science – HarvardX (edX)
- CS50P: Introduction to Programming with Python – HarvardX (edX)
- AWS Academy Graduate – Cloud Developing – Training Badge
- AWS Academy Graduate – Cloud Foundations – Training Badge